

LAB 6 – SORTING (1230519)

① Insertion

9, 5, 20, 1, 4, 3, 2, 7, 13, 25

5, 9, 20, 1, 4, 3, 2, 7, 13, 25

1, 5, 9, 20, 4, 3, 2, 7, 13, 25

1, 4, 5, 9, 20, 3, 2, 7, 13, 25

1, 3, 4, 5, 9, 20, 2, 7, 13, 25

1, 2, 3, 4, 5, 9, 20, 7, 13, 25

1, 2, 3, 4, 5, 7, 9, 20, 13, 25

1, 2, 3, 4, 5, 7, 9, 13, 20, 25

Selection (increasing order)

9	5	20	1	4	3	2	7	13	25
1	5	20	9	4	3	2	7	13	25
1	2	20	9	4	3	5	7	13	25
1	2	3	9	4	20	5	7	13	25
1	2	3	4	9	20	5	7	13	25
1	2	3	5	9	20	7	13	25	
1	2	3	4	5	7	9	20	13	25
1	2	3	4	5	7	9	13	20	25

Bubble

9	5	20	1	4	3	2	7	13	25
5	9	20	1	4	3	2	7	13	25
5	9	20	1	4	3	2	7	13	25
5	9	1	20	4	3	2	7	13	25
5	9	1	4	20	3	2	7	13	25
5	9	1	4	3	20	2	7	13	25
5	9	1	4	3	2	20	7	13	25
5	9	1	4	3	2	7	20	13	25
5	9	1	4	3	2	7	13	20	25
5	1	9	4	3	2	7	13	20	25
5	1	4	9	3	2	7	13	20	25
5	1	4	3	9	2	7	13	20	25
5	1	4	3	2	9	7	13	20	25
5	1	4	3	2	7	9	13	20	25
1	5	4	3	2	7	9	13	20	25
1	4	5	3	2	7	9	13	20	25
1	4	3	5	2	7	9	13	20	25
1	4	3	2	5	7	9	13	20	25
1	3	4	2	5	7	9	13	20	25
1	3	2	4	5	7	9	13	20	25
1	2	3	4	5	7	9	13	20	25

Merge Sort

9 5 20 1 4 3 2 7 13 25

9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25
9	5	20	1	4	3	2	7	13	25

5 9 1 4 20 2 3 7 13 25

1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25
1	4	5	9	20	2	3	7	13	25

1 2 3 4 5 7 9 13 20 25

Radix												
9	5	20	1	4	3	2	7	13	25			
20	1	2	3	4	5	6	7	8	9			
= [20, 1, 2, 3, 13, 4, 25, 7, 9]												
9												
7												
5												
4												
2												
1												
0												
= [1, 2, 3, 4, 5, 7, 9, 13, 20, 25]												
② Insertion :												
9	5	20	1	4	3	2	7	13	25			
Pass 1 : compare 5 and 9 stop												
9	5	20	1	4	3	2	7	13	25			
Pass 2 : compare 20 and 5 swap												
compare 20 and 9 swap												
20	9	5	1	4	3	2	7	13	25			
Pass 3 : compare 4 and 1 swap												
compare 4 and 5 stop												
20	9	5	4	1	3	2	7	13	25			
Pass 4 : compare 3 and 1 swap												
compare 2 and 3 stop												
20	9	5	4	3	1	2	7	13	25			
Pass 5 : compare 2 and 1 swap												
compare 2 and 3 stop												
20	9	5	4	3	2	1	7	13	25			
Pass 6 : compare 7 and 1 swap												
compare 7 and 2 swap												
compare 7 and 3 swap												
compare 7 and 4 swap												
compare 7 and 5 swap												
compare 7 and 9 stop												
20	9	7	5	4	3	2	1	13	25			
Pass 7 : compare 13 and 1 swap												
compare 13 and 2 swap												
compare 13 and 3 swap												
compare 13 and 4 swap												
compare 13 and 5 swap												
compare 13 and 7 swap												
compare 13 and 9 swap												
compare 13 and 20 stop												
20	13	9	7	5	4	3	2	1	25			
Pass 8 : compare 25 and 1 swap												
compare 25 and 2 swap												
compare 25 and 3 swap												
compare 25 and 4 swap												
compare 25 and 5 swap												
compare 25 and 7 swap												
compare 25 and 9 swap												
compare 25 and 13 swap												
compare 25 and 20 swap												
25	20	13	9	7	5	4	3	2	1			
Pass 9 : compare 25 and 20 stop												
25	20	13	9	7	5	4	3	2	1			

Selection Sort

9	5	20	1	4	3	2	7	13	25
---	---	----	---	---	---	---	---	----	----

Pass 1

compare 9 and 25 swap

25	5	20	1	4	3	2	7	13	9
----	---	----	---	---	---	---	---	----	---

Page 2

compare 5 and 20 jump

25	20	5	1	4	3	2	7	13	9
----	----	---	---	---	---	---	---	----	---

Page 3

compare 5 and 12 swap

25	20	19	1	4	2	2	7	5	9
----	----	----	---	---	---	---	---	---	---

Page 24

compare 1 and 9 swap

25	20	13	9	4	3	2	7	5	1
----	----	----	---	---	---	---	---	---	---

Pass 5

compare 4 and 7 row

25	20	13	9	7	3	2	4	5	1
----	----	----	---	---	---	---	---	---	---

loss 6

compare 3 and 5 swap

25	20	13	9	7	5	2	4	3	1
----	----	----	---	---	---	---	---	---	---

Page 7

compare 14 and 2 swap

25 20 13 9 7 5 4 2 3 1

Pass 8

compare 3 and 2 swap

25	20	13	9	7	5	4	3	2	1
----	----	----	---	---	---	---	---	---	---

Pass 9

compare 2 and 1 swap

25	20	13	9	7	5	4	3	2	1
----	----	----	---	---	---	---	---	---	---

Bubble sort

9	5	20	1	4	3	2	7	13	25
---	---	----	---	---	---	---	---	----	----

25	9	5	20	1	4	3	2	7	13
----	---	---	----	---	---	---	---	---	----

25	9	5	20	17	1	4	3	2	7
----	---	---	----	----	---	---	---	---	---

25	9	5	20	13	7	1	4	3	2
----	---	---	----	----	---	---	---	---	---

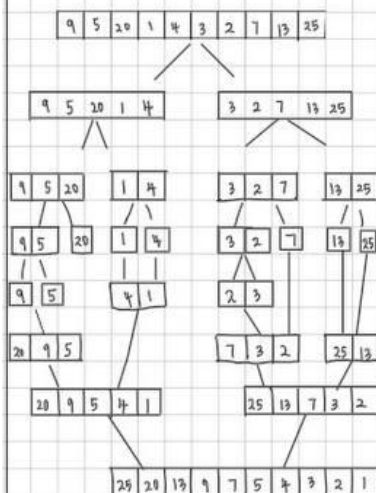
25	9	6	30	18	7	4	1	3	2
----	---	---	----	----	---	---	---	---	---

25	9	5	20	13	7	4	3	2	1
----	---	---	----	----	---	---	---	---	---

25	20	9	5	13	7	4	3	2	1
----	----	---	---	----	---	---	---	---	---

25	20	13	9	5	7	4	3	2	1
----	----	----	---	---	---	---	---	---	---

35	20	13	9	7	5	4	3	2	1
----	----	----	---	---	---	---	---	---	---



Radix											
	9	5	20	1	4	3	2	7	13	25	
					13		25				
		20	1	2	3	4	5		7	9	
queue	0	1	2	3	4	5	6	7	8	9	
data =	[20, 1, 2, 3, 13, 4, 5, 25, 7, 9]										
		9									
		7									
		5									
		4									
		3									
		2		25							
		1	13	20							
queues		0	1	2	3	4	5	6	7	8	9
data =	[25, 20, 13, 9, 7, 5, 4, 3, 2, 1]										

3. InsertionSort(A)

for $i = 1$ to $\text{length}(A) - 1$ do:

$\text{key} = A[i]$

$j = i - 1$

 while $j \geq 0$ and $A[j] > \text{key}$ do:

$A[j + 1] = A[j]$

$j = j - 1$

$A[j + 1] = \text{key}$

SelectionSort(array):

for i from 0 to $\text{length}(\text{array}) - 1$:

$\text{minIndex} = i$

 for j from $i + 1$ to $\text{length}(\text{array}) - 1$:

 if $\text{array}[j] < \text{array}[\text{minIndex}]$:

$\text{minIndex} = j$

 if $\text{minIndex} \neq i$:

$\text{swap}(\text{array}[i], \text{array}[\text{minIndex}])$

```
BubbleSort(array):  
  for i from 0 to length(array) - 1:  
    for j from 0 to length(array) - i - 2:  
      if array[j] > array[j + 1]:  
        swap(array[j], array[j + 1])
```